

HWK 5
ADVANCED MACHINE LEARNING
DATA 642

Objective: This homework aims to enhance students' practical and theoretical understanding of clustering algorithms and Gaussian mixture models through hands-on exercises with real-world and simulated datasets. The first exercise involves working with the Olivetti faces dataset, where students will load, preprocess, and split the data into training, validation, and test sets. Using stratified sampling to maintain an equal number of images per person across sets, they will then apply K -Means clustering to the images, visualize the clusters, and interpret the results to see if similar faces are grouped together. The second exercise involves generating data points from two-dimensional Gaussian distributions and applying the Gaussian Mixture Model (GMM) with Expectation-Maximization (EM) to estimate distribution parameters. Students will sample data, run the EM algorithm, and visualize clustering results, experimenting with different values of K and initialization points to observe their impact on the results. Through these exercises, students will gain insights into the application and performance of clustering and mixture models, enhancing their ability to analyze and interpret high-dimensional data.

Exercise 1 (5 points)

The classic Olivetti faces dataset contains 400 grayscale 64×64 pixel images of faces. Each image is flattened to a 1D vector of size 4,096. 40 different people were photographed (10 times each), and the usual task is to train a model that can predict which person is represented in each picture. Load the dataset using the `sklearn.datasets.fetch_olivetti_faces()` function, then split it into a training set, a validation set, and a test set (note that the dataset is already scaled between 0 and 1). Since the dataset is quite small, you probably want to use stratified sampling to ensure that there are the same number of images per person in each set. Next, cluster the images using K -Means, and ensure that you have a good number of clusters (using one of the techniques discussed in this chapter). Visualize the clusters: do you see similar faces in each cluster?

Exercise 2 (10 points)

Generate 100 data points from each of the two-dimensional Gaussian distributions with parameters

$$\boldsymbol{\mu}_1^\top = \begin{bmatrix} 0.9 & 1.02 \end{bmatrix}^\top, \quad \boldsymbol{\mu}_2^\top = \begin{bmatrix} -1.2 & -1.3 \end{bmatrix}^\top$$

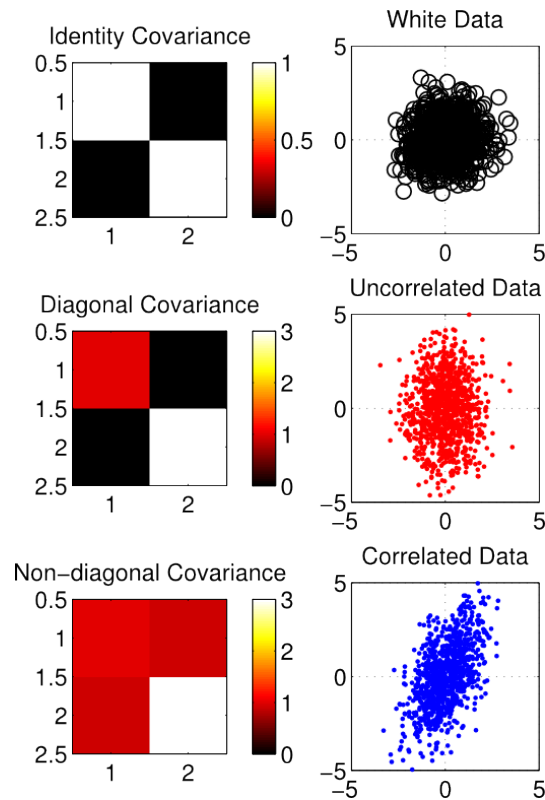
and

$$\boldsymbol{\Sigma}_1^\top = \begin{bmatrix} 0.5 & 0.081 \\ 0.081 & 0.7 \end{bmatrix}, \quad \boldsymbol{\Sigma}_2^\top = \begin{bmatrix} 0.4 & 0.02 \\ 0.02 & 0.3 \end{bmatrix}$$

- (a) Sample $N_1 = 100$ and $N_2 = 20$ points from each distribution and run Gaussian mixtures EM to estimate the parameters of the distribution. Plot your clustering results of the EM algorithm as well as the results of just the initial values of the parameters.
- (b) Try different values for K . Comment on the results.

- (c) Play with different initialization points and also try points which are too far from the true mean values of the clusters. Comment on the results.

Note: Correlation between variables in a l -dimensional dataset are often summarized by an $l \times l$ covariance matrix. To get a better understanding of how correlation matrices characterize correlations between data points, we plot data points drawn from 3 different 2-dimensional Gaussian distributions, each of which is defined by a different covariance matrix.



The left plots below display the 2×2 covariance matrix for each Gaussian distribution. The values along the diagonal represent the variance of the data along each dimension, and the off-diagonal values represent the covariances between the dimensions. Thus the i, j -th entry of each matrix represents the correlation between the i -th and j -th dimensions. The right plots show data drawn from the corresponding 2D Gaussian. For more info visit (<https://theclevermachine.wordpress.com/2013/03/29/covariance-matrices-and-data-distributions/>)